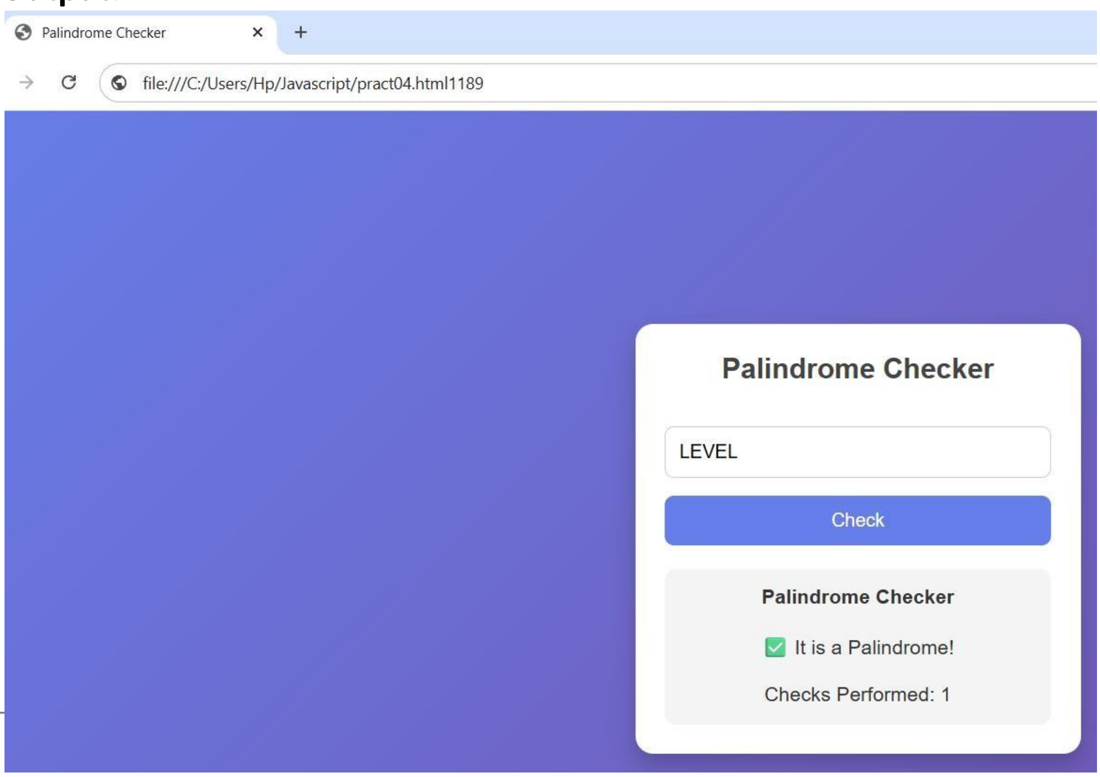
Experiment No.: 04

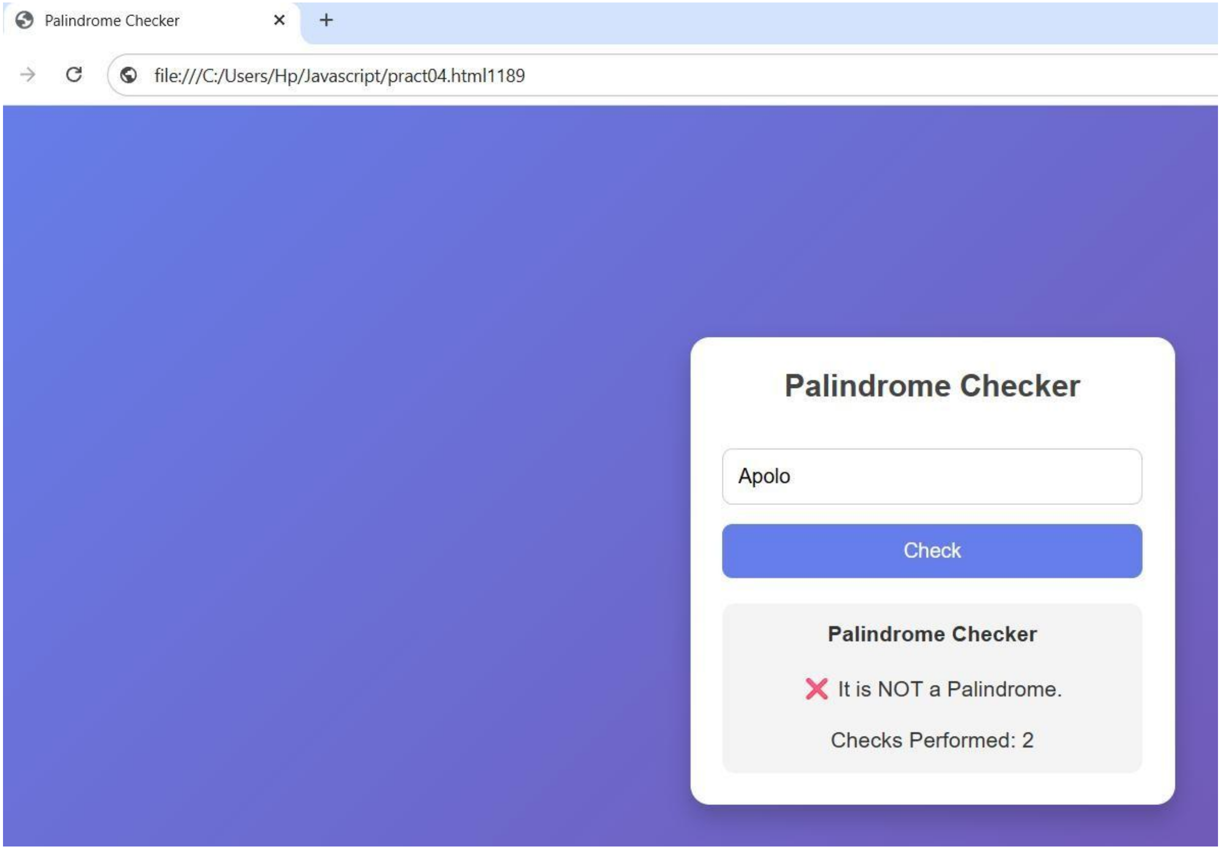
1. **Title:** Use function types, scope, and closures; apply try-catch; build a palindrome checker.

2. **Software/Tools Required:** HTML, JavaScript, CSS, VS Code.

3. **Theory:** This program implements a Palindrome Checker using HTML, CSS, and JavaScript. It allows the user to enter a word or sentence and checks whether it is a palindrome. The program demonstrates different types of functions, including a function declaration, function expression, and arrow function. It also illustrates variable scope by using both global and local variables. A closure is implemented to count the number of palindrome checks performed while preserving the counter value between function calls. Exception handling is achieved using try...catch, which displays an appropriate error message if the input is empty. The webpage dynamically displays the result without refreshing the page.

4. **Output:**




Palindrome Checker
file:///C:/Users/Hp/Javascript/pract04.html1189
Palindrome Checker
Apolo
Check
Palindrome Checker
❌ It is NOT a Palindrome.
Checks Performed: 2

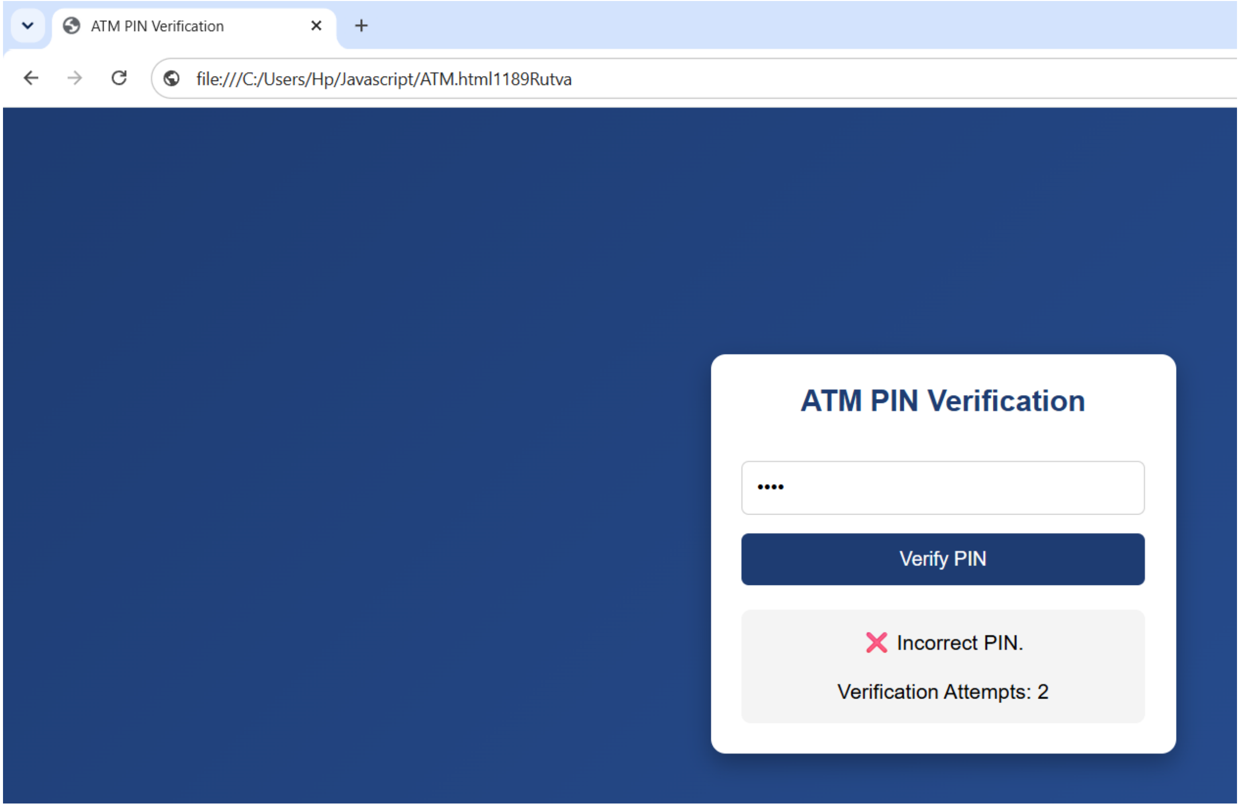
CASE STUDY 04

Case Study Title: To implement an ATM Card PIN Verification system using JavaScript by demonstrating function declaration, function expression, arrow functions, variable scope, closures, and exception handling.

Code:

```
Welcome | Practical03.html | pract04.html | department.html | billing.html | inde
ATM.html > html
<html lang="en">
<body>
<script>

// Global Scope
let storedPIN = "1234";

// Function Declaration
function createCounter(){
    let attempts = 0;

    // Closure
    return function(){
        attempts++;
        return attempts;
    };
}

const counter = createCounter();

// Function Expression
const isValidPIN = function(pin){
    return /^\d{4}$/.test(pin);
};

// Arrow Function
const checkPIN = pin => pin === storedPIN;

// Main Function
function verifyPIN(){

    try{

        let pin = document.getElementById("pin").value;

        if(pin=="")
            throw "Please enter your PIN.";

        if(!isValidPIN(pin))
            throw "PIN must contain exactly 4 digits.";

        let message;

        if(checkPIN(pin))
            message = "✅ PIN Verified Successfully!";
        else
            message = "❌ Incorrect PIN.";

        document.getElementById("result").innerHTML =
        message +
        "<br><br>Verification Attempts: " + counter();

    }

    catch(error){

        document.getElementById("result").innerHTML =
        "<span style='color:red'>" + error + "</span>";

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
```

Output:



Result/Conclusion: The Palindrome Checker successfully demonstrates important JavaScript concepts such as function types, variable scope, closures, and exception handling (try...catch). It validates user input, processes the entered text by removing spaces and special characters, and determines whether it is a palindrome. The project provides a user-friendly interface and showcases how JavaScript can be used to build interactive, reliable, and efficient web applications.